

STAT X12: Homework 4

NAME(S)

Due, Friday 4/24/26, 11:59pm

Instructions

You may complete this assignment individually or in a group of up to 4 people of your choice. Please turn in one assignment per group and associate all participants names with your submission. For code chunks with code, work through each code chunk line by line and then change `eval = FALSE` to `eval = TRUE` when you understand the code and it runs without errors.

Sitting and Back pain

Gupta *et al.* (2015) found a positive association between sitting time and higher levels of low back pain in a specific group of blue-collar workers in Denmark. The primary analysis was done using binary logistic regression with low back pain/not as a binary response variable (0/1: $Y_i = 1$ if back pain was reported from participant i and 0 if not).

We will consider RQs associated with two of the author's models: ("Crude" model) "*How does sitting time relate to the odds of high back pain for individuals in this population*", ("Step 1" model) "*How does sitting time relate to odds of high low back pain for individuals in this population of similar age, sex, smoking status and bmi (i.e., after accounting for age, sex, smoking and bmi)?*"

We have some data cleaning to do. Be sure to work through the code in the next code chunk line-by-line so you can see what each line does.

```
# note packages are loaded above
# read in the data
gupta_2015 <- read_csv("gupta_2015_sheet1.csv")

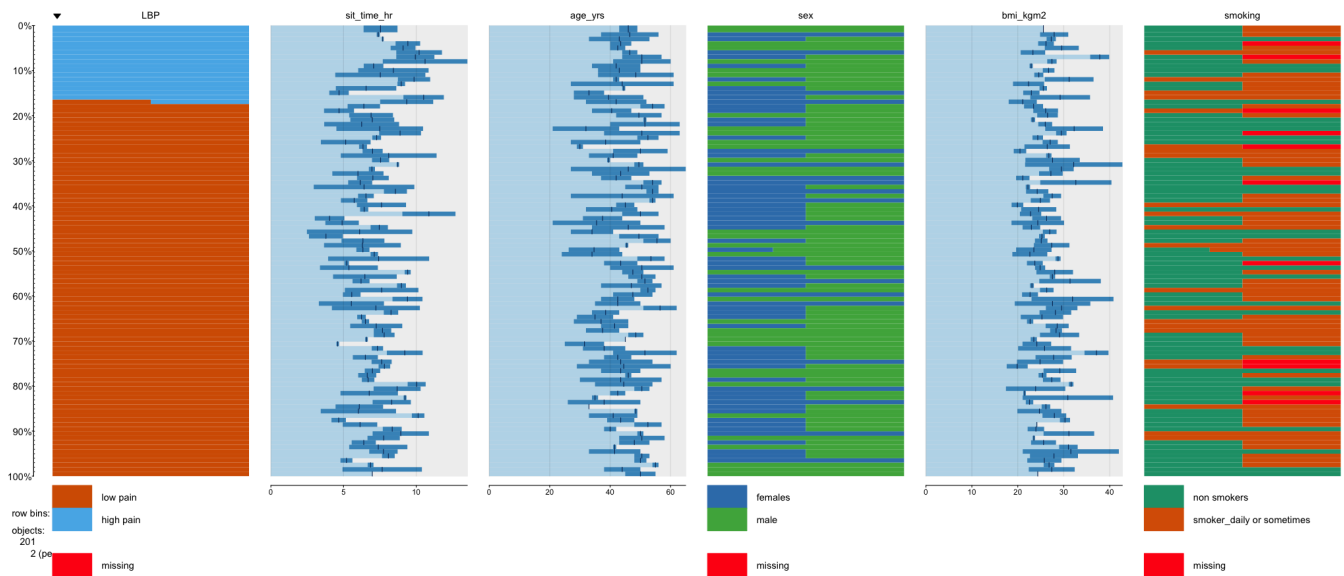
# clean data.
df_gupta <- gupta_2015 |>
  # select the variables we will use for the analysis all for RQ2
  dplyr::select("age in years",
               "sex",
               "body mass index in kg/m2",
               "total sitting time in hours",
```

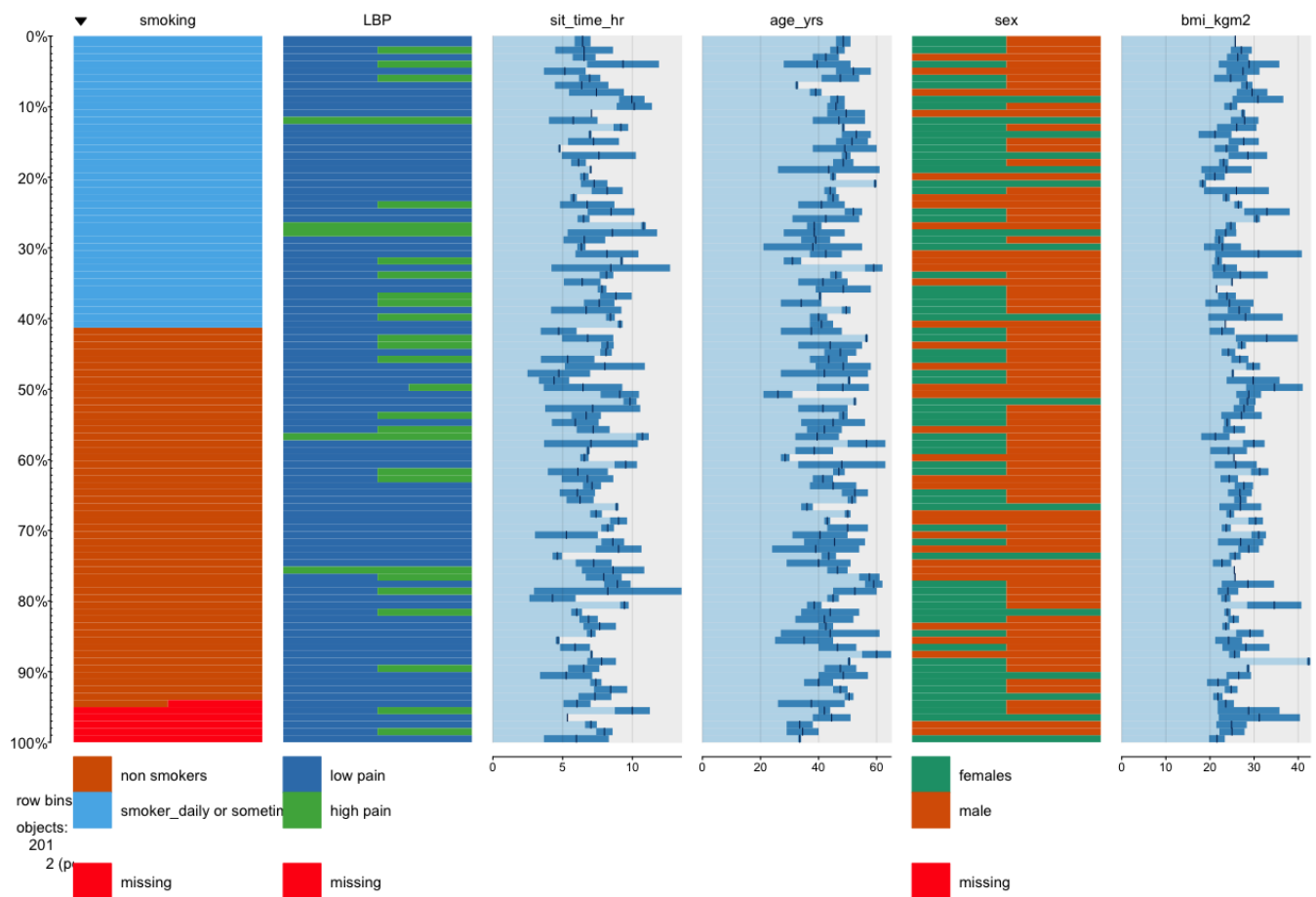
```

    "smoking status categories",
    "low back pain categories") |>
# create shorter variables names
rename(age_yrs = "age in years",
       bmi_kgm2 = "body mass index in kg/m2",
       sit_time_hr = "total sitting time in hours",
       smoking = "smoking status categories",
       LBP = "low back pain categories") |>
# create factors and make sure low pain = 0 and high pain = 1 for LBP
mutate(sex = factor(sex),
       smoking = factor(smoking),
       LBP = factor(LBP, levels = c("low pain", "high pain")),
       high_pain = ifelse(LBP == "high pain", 1, 0))
glimpse(df_gupta)

```

Now let's visualize these data with a few tabplots. The .png files are provided on Canvas for inclusion in this document and for your reference.





1. (5pts) Tabplots can be useful for visualizing patterns between the response and other variables of interest (Tabplot 1), and to investigate if assuming missing values are missing at random is a “visually reasonable” assumption (Tabplot2). Use the tabplots to comment on the association between sit time and LBP and the “reasonableness” of removing NA values from the dataset before analysis. *Hint: think about if there are any patterns in the variables that are systematically different for complete cases (cases with values for all variables) and incomplete cases.* Make reference to the plots in your answer to this question.

The following code chunk removes cases with missing smoking status from `df_gupta`. This makes the assumption that missing values are missing completely at random, which is something we would want to justify with more than just reference to a Tabplot. The researchers could help with this.

```
# remove NAs
df_final <- df_gupta |>
  filter(!is.na(smoking))
# check that there are no NAs, sum should be 0 if all evaluations are FALSE
sum(is.na(df_final))
```

2. (5 pts) Create a `cdplot` for LBP vs. `sittime`. Describe what this plot tells you about the marginal (overall) relationship between sit time and low back pain.

3. **(5 pts)** The following code will create empirical logit plots for all continuous variables in this study. Does the assumption of linearity on the logit scale seem appropriate for these variables? (Assuming an additive model we can look at each of the marginal relationships to address this). How could [Katie's] choice to split the quantitative X variables into 5 groups impact your answer to this question? *Note: You can play around with different #s of groups by changing the **breaks** argument in the code.*
4. **(5pts)** Write out a theoretical model to investigate the crude model research question, *How does sitting time relate to the odds of high back pain for individuals in this population?* Define the response variable, any indicator variables, and identify the parameter of interest.
5. **(10 pts)** Write code to estimate the model in 4 and conduct a Wald-Based test for addressing the research question. Provide a complete summary of statistical findings, and an effects plot on the transformed scale.
6. **(5 pts)** Write code to generate a 95% profile-likelihood based CI for the parameter of interest in (4). Interpret the interval in the context of the problem and on the odds scale.
7. **(5pts)** How do results and the sample size of the data set you used in 5 and 6 compare to Gupta 2015's reported point estimate and CI for their "Crude" model (can be found in their Table 3)? Explain how you could clean the data differently to resolve this discrepancy (you do not need to do this, just explain how you could). *Note that, with a quantitative predictor, the odds ratio (OR) is the estimated multiplicative change in the odds of a success for a one unit increase in the predictor (see notes from this week for another example of this).*
8. **(5pts)** Write out a theoretical model to investigate the crude model research question, *How does sitting time relate to odds of high low back pain for individuals in this population of similar age, sex, smoking status and bmi?* Define the response variable, any indicator variables, and identify the parameter of interest.
9. **(10 pts)** Write code to estimate the model in 8 and conduct a LRT for addressing the research question. Provide the estimated model and a complete summary of statistical findings.
10. **(5 pts)** Write code to generate a 95% profile-likelihood based CI for the parameter of interest in (4). Interpret the interval in the context of the problem and on the odds scale.
11. **(5pts)** How do your results compare to the author's findings for their "Step 1" model? Now you should have the same sample size. What else could describe any slight discrepancies in the results of the CI reported in the paper and yours?